

The basic of linearization and wave ansatz in fluid mechanics



Marek Stastna

Motivation

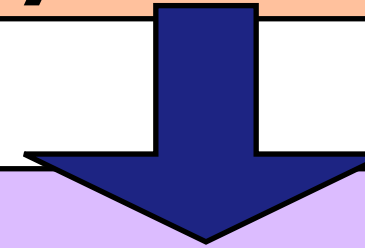
- ❖ Fluid Mechanics is an old subject meaning that the recent period of being able to find numerical solution is not representative of the historically developed understanding.
- ❖ Some important exact solutions are available, but it is fair to say that the bulk of historical work has been ‘by approximation’.
- ❖ In these slides we introduce a key approximation technique, linearization, and show it in action.
- ❖ We then discuss a number of applications.
- ❖ In this slide deck we first consider the question of flow stability, or do perturbations to a flow make it evolve and change fundamentally or do they die away, leaving the original flow unchanged.
- ❖ We then derive the linear description of waves in a stratified wave guide like the coastal ocean.

Basic Set up I

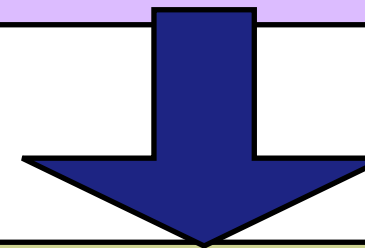
- ❖ The lab book activities for this slide deck are largely related to algebra.
- ❖ The work cycle would consist of paper and pen work, some proof reading and subsequent latex work to write up what you derived.
- ❖ There are many linearization exercises in classical fluid mechanics texts.

The typical structure of analytical theory for fluid mechanics

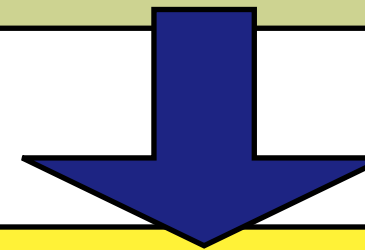
The Navier Stokes or Euler equations
(highly nonlinear)



Choose suitable background state to linearize around



Write solution as background + small perturbation
Perform algebra to find linearized equations



Use classical methods for linear PDEs to solve or
analyze the linearized equations

Constant density Euler Equations

Momentum

$$\frac{D\vec{u}}{Dt} = -\frac{1}{\rho_0} \nabla p - g\hat{k}$$

Mass

$$\vec{\nabla} \cdot \vec{u} = 0$$

In AMATH 361 we discussed how to reduce these equations to Laplace's equation.
This was called potential flow, or flow of inviscid, incompressible, irrotational fluids.
This can be a good topic for a focused lab book exploration.

Absorbing pressure: the one simple balance we can form in the Euler equations

$$0 = -\frac{\partial p_H}{\partial z} - \rho_0 g \quad \text{hydrostatic balance}$$

$$\text{Let, } p = p_H + p_D$$

$$\rho_0 \frac{D\vec{u}}{Dt} = -\nabla p_H - \nabla p_D - \rho_0 g \hat{k}$$


these balance

$$\boxed{\rho_0 \frac{D\vec{u}}{Dt}} = \boxed{-\nabla p_D}$$

And we usually drop the subscript D

$$\boxed{\vec{\nabla} \cdot \vec{u} = 0}$$

Conservation of mass,
just for completeness

Vorticity in 2D

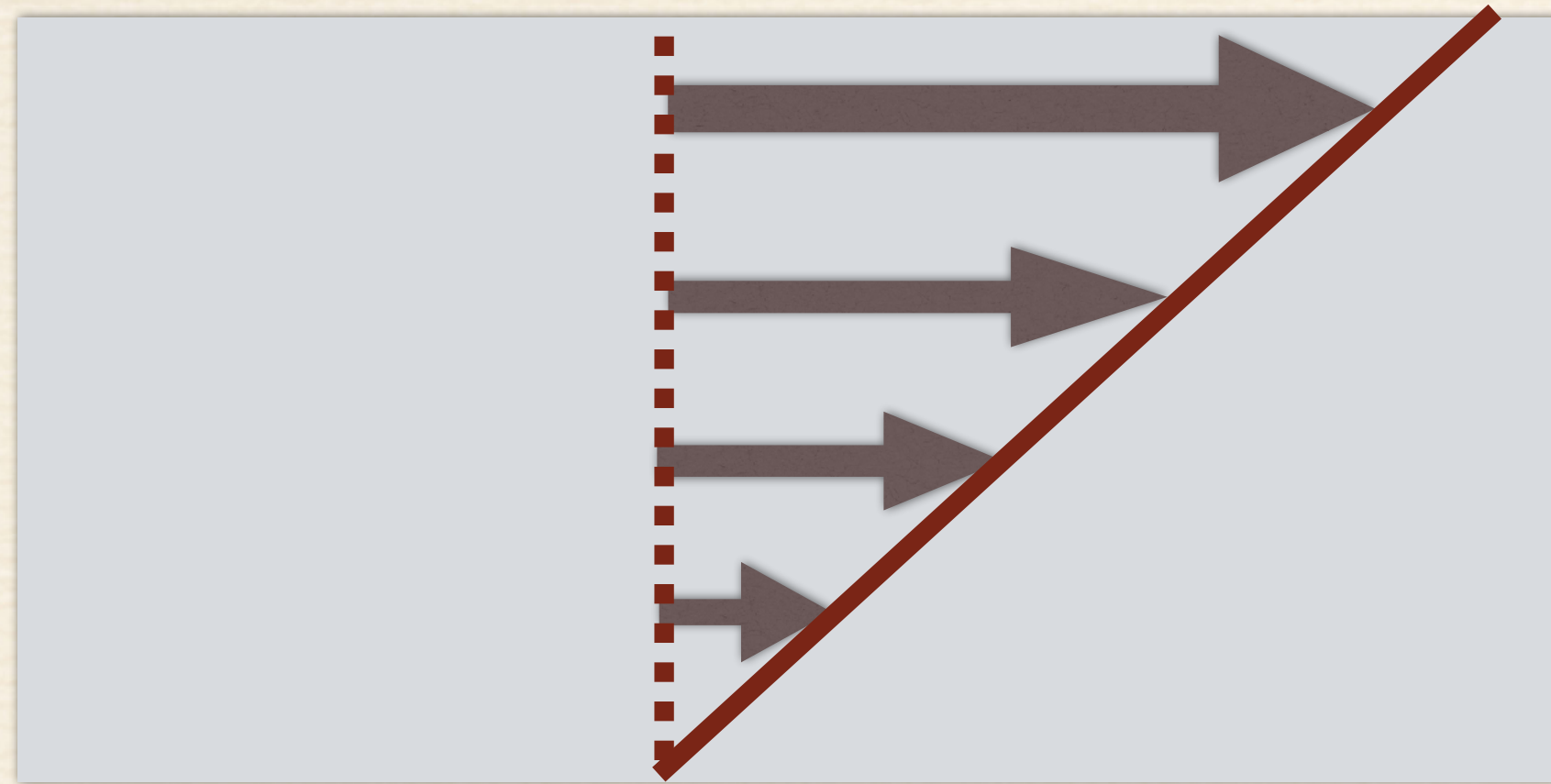
- ❖ Consider a 2D flow $\vec{u} = (u(x, z, t), 0, w(x, z, t))$.
- ❖ The vorticity in this case (subscripts denote partial derivatives) reads:
$$\vec{\omega} = (0, \omega(x, z, t), 0) = (0, u_z - w_x, 0)$$
- ❖ Cross-differentiating gives the vorticity equation which says vorticity is conserved following a fluid particle:
$$\frac{D\omega}{Dt} = 0 \implies \omega_t + u\omega_x + w\omega_z = 0$$

Linearized Vorticity in 2D

- ❖ Consider a 2D flow split as follows $\vec{u} = (U(z), 0) + \epsilon(u(x, z, t), w(x, z, t))$ where $\epsilon \ll 1$.
- ❖ $U(z)$ is called a shear flow and you can confirm that when there is no pressure gradient, the Euler equations allow for ANY $U(z)$ to be a solution.
- ❖ The vorticity in this case (subscripts denote partial derivatives) reads:
$$\omega^{total}(x, z, t) = U_z + \epsilon(u_z - w_x) = \Omega(z) + \epsilon\omega(x, z, t).$$
- ❖ Cross-differentiating gives the vorticity equation which says vorticity is conserved following a fluid particle, but the introduction of epsilon changes things slightly:
$$\frac{D\omega^{total}}{Dt} = 0 \implies \epsilon(\omega_t + U\omega_x + w\Omega_z) + \epsilon^2 w\omega_z = 0 \approx \omega_t + U\omega_x + w\Omega_z = 0$$

In linearization we drop the epsilon squared terms. This is regular perturbation theory

Linear shear, the simplest case.

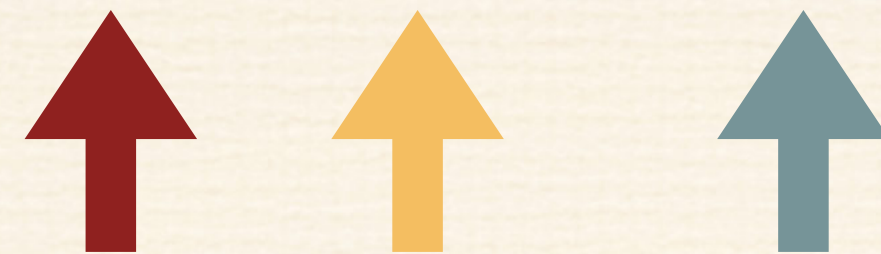


$$U(z) = \gamma z \implies \Omega = \gamma$$

- ❖ There are many shear profiles possible.
- ❖ It will turn out even the simplest precludes closed form solutions (a common story in fluid mechanics)
- ❖ But the upshot is that shear flows, even simple ones have vorticity associated with them

Linearized Vorticity Equation: mechanisms

$$\omega_t + U\omega_x + w\Omega_z = 0$$



- ❖ Rate of change with time at a fixed location.
- ❖ Advection by the “base flow” $U(z)$.
- ❖ Exchange with the background vorticity Ω
- ❖ Can rewrite as a linearized material derivative and a “source/sink” term.
- ❖ The idea is that for different choices of $U(z)$ the interaction may be different.

$$\frac{D_{lin}\omega}{Dt} + w\Omega_z = 0$$

A bit of Background: The Stream Function

In 2D, say (x,z), the conservation of mass allows use to make a significant simplification.

$$\vec{\nabla} \cdot \vec{u} = 0$$

$$\vec{u} = (u(x, z, t), w(x, z, t))$$

Vector calculus then guarantees the existence of a function called the stream function, from which the velocity components may be calculated as:

$$(u(x, z, t), w(x, z, t)) = \left(\frac{\partial \psi}{\partial z}, -\frac{\partial \psi}{\partial x} \right)$$

This also means that the stream lines are the level curves of psi or the tangent curves of the velocity field and the expression for vorticity is simplified

$$\psi(x, z, t) = C$$

$$\omega = u_z - w_x = \psi_{zz} + \psi_{xx}$$

Linearized Vorticity Equation: wave ansatz

$$\omega_t + U\omega_x + w\Omega_z = 0 \implies (\nabla^2\psi)_t + U\nabla^2\psi_x - \psi_x U_{zz} = 0$$

$$\psi = \text{Re} [\exp(i(kx - \sigma t))\phi(z)]$$

$$\nabla^2\psi = \text{Re} [(\phi_{zz} - k^2\phi)\exp(i(kx - \sigma t))]$$

$$[-i\sigma + ikU](\phi_{zz} - k^2\phi) - ikU_{zz}\phi = 0$$

I have suppressed the common factor
of $\exp(i(kx - \sigma t))$

$$[\sigma - kU](\phi_{zz} - k^2\phi) + kU_{zz}\phi = 0$$

$$[U - c](\phi_{zz} - k^2\phi) - U_{zz}\phi = 0$$

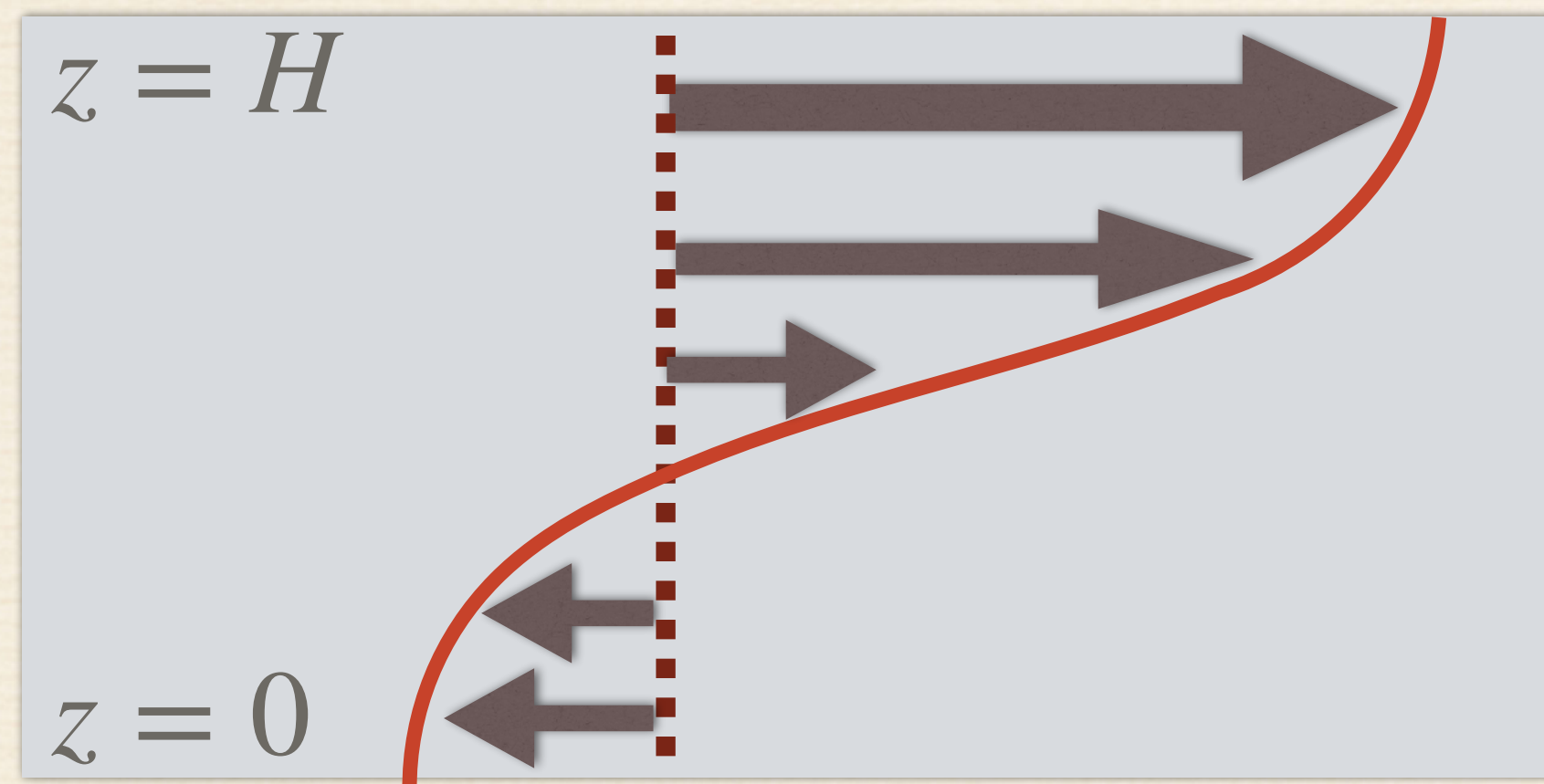
Let $c = \sigma/k$

- ❖ This calculation has reduced a PDE for vorticity to an ODE
- ❖ The ODE is complicated by the fact that we need to solve for “c” along with phi (i.e. an eigenvalue problem).

More about the wave ansatz

- ❖ The solution form $\psi = \text{Re} [\exp(i(kx - \sigma t))\phi(z)]$ is called the “wave ansatz”, or sometimes the plane wave solution (this is easier to see in 3D than in 2D).
- ❖ Let's assume that k , the wave number, is a known, measured parameter.
- ❖ Let's also rewrite the complex exponential as $\exp(ik(x - ct))$ and let's consider a few cases:
- ❖ If “ c ” is real then $\text{Re}[\exp(ik(x - ct))] = \cos(k(x - ct))$ which is a traveling cosine moving from left to right (when $c > 0$) with a fixed amplitude of 1.
- ❖ When $c = c_R \pm ic_I$ we have $\text{Re}[\exp(ik(x - ct))] = \cos(k(x - c_R t))\exp(\pm kc_I t)$ and now the real part of “ c ” gives a traveling cosine while the imaginary part gives an envelope that is either growing or shrinking in time.
- ❖ Since complex “ c ” come in conjugate pairs (because the coefficients of the ODE are real) whenever “ c ” complex we get a solution for ψ that grows in time.

Rayleigh's Stability Equation



$$[U - c](\phi_{zz} - k^2\phi) - U_{zz}\phi = 0$$

$$w = 0 \implies \psi_x = 0 \implies \phi(0) = \phi(H) = 0$$

$$c_I \neq 0 \implies \text{perturbations grow}$$

- ❖ Consider a general shear flow pictured above.
- ❖ If one can solve the so-called Rayleigh's equation for c and ϕ , then one can determine whether small perturbations to $U(z)$ (which occur all the time in the real world) will grow in time.
- ❖ If they do grow in time, such a flow is called unstable and probably won't be observed in an experiment.
- ❖ Hydrodynamic stability theory is one of the most mathematical sub-branches of fluid mechanics with a long history.
- ❖ Typically some numerical methods are needed, but these are easily accessible even on laptops.

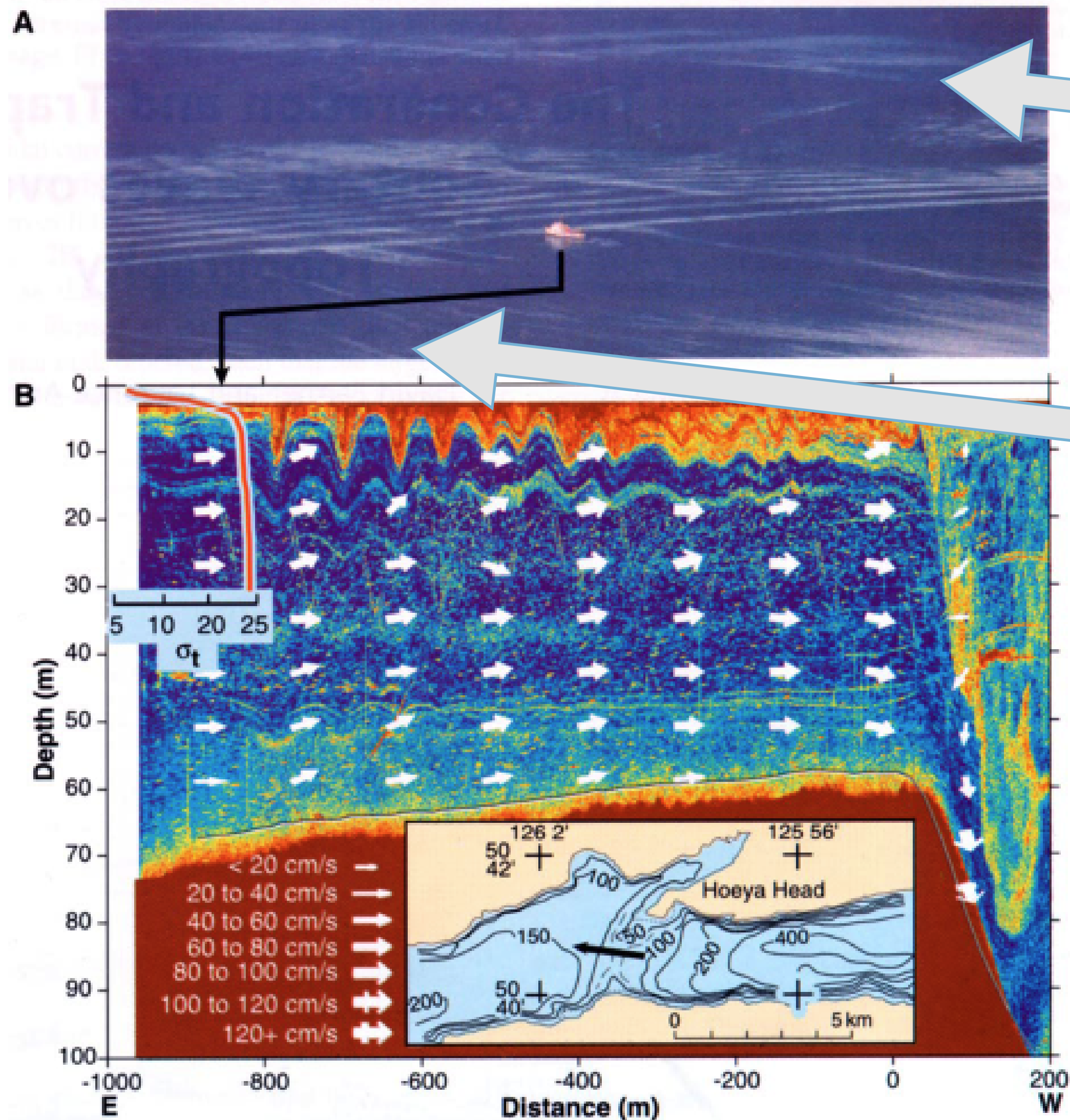
The presence of a stable density stratification is the fundamental physical property of both the atmosphere and natural bodies of water on scales ranging from a meter or so to a kilometre or more (before the Earth's rotation begins to play a strong role).



In this picture of the Olympic mountains across from the Strait of Juan de Fuca from Victoria, B.C. the well established cloud layers indicate a stably stratified situation. Due to stratification clouds cannot just go up and over the mountains and remain stuck in clear bands.

Stable density stratification allows for a class of waves for which the restoring force is gravity. These can be observed in the ocean, the atmosphere and the laboratory.

Internal
Solitary-Like
Waves in
Knight Inlet



**SURFACE
MANIFESTATIONS
OF INTERNAL
WAVES**

Actual Doppler
signatures of
internal waves

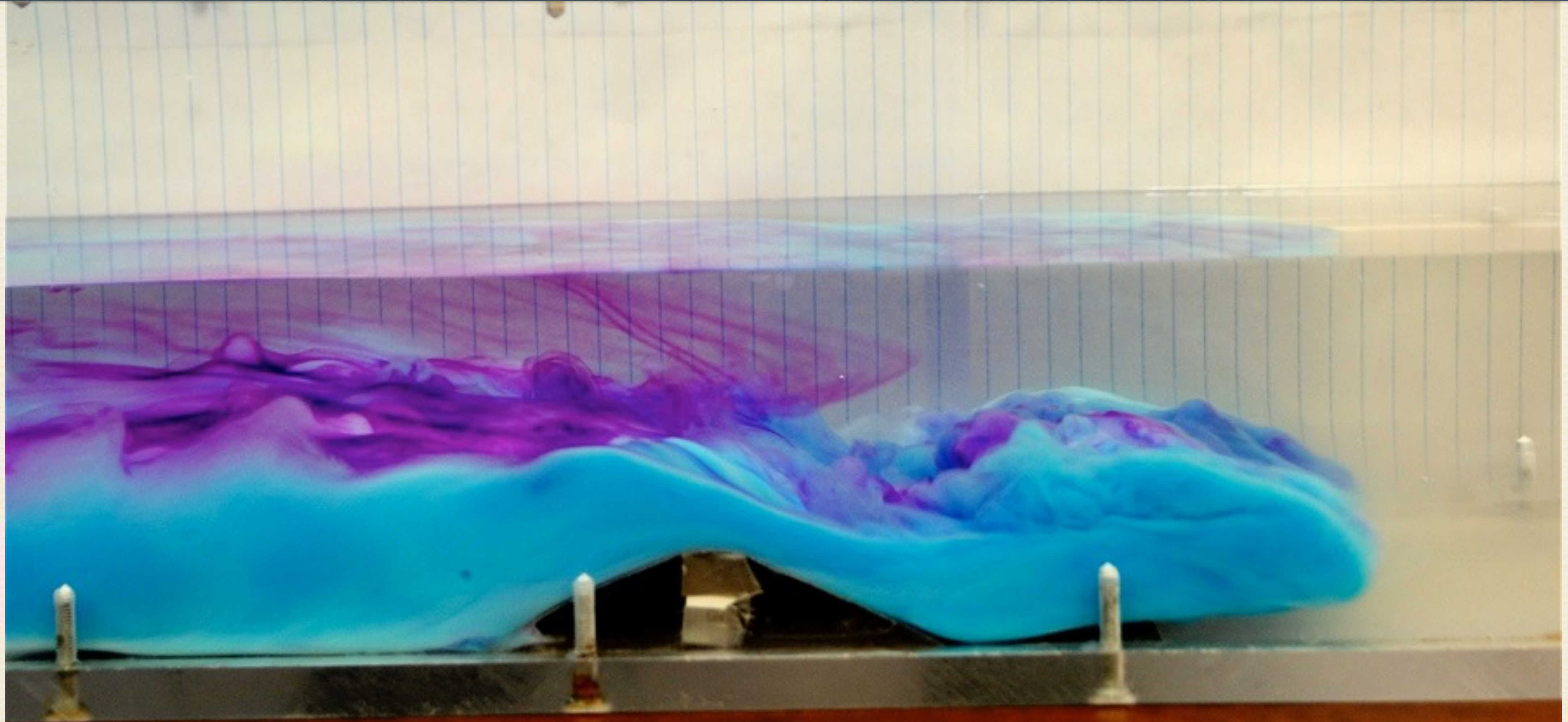
breaking lee
wave; still a bit
of a mystery



Morning Glory Clouds, Gulf of Carpentaria, Australia (wikipedia)



In the lab, the easiest density stratified phenomenon to create is a gravity current, or an intrusion of heavy (blue) fluid into light (clear) fluid.



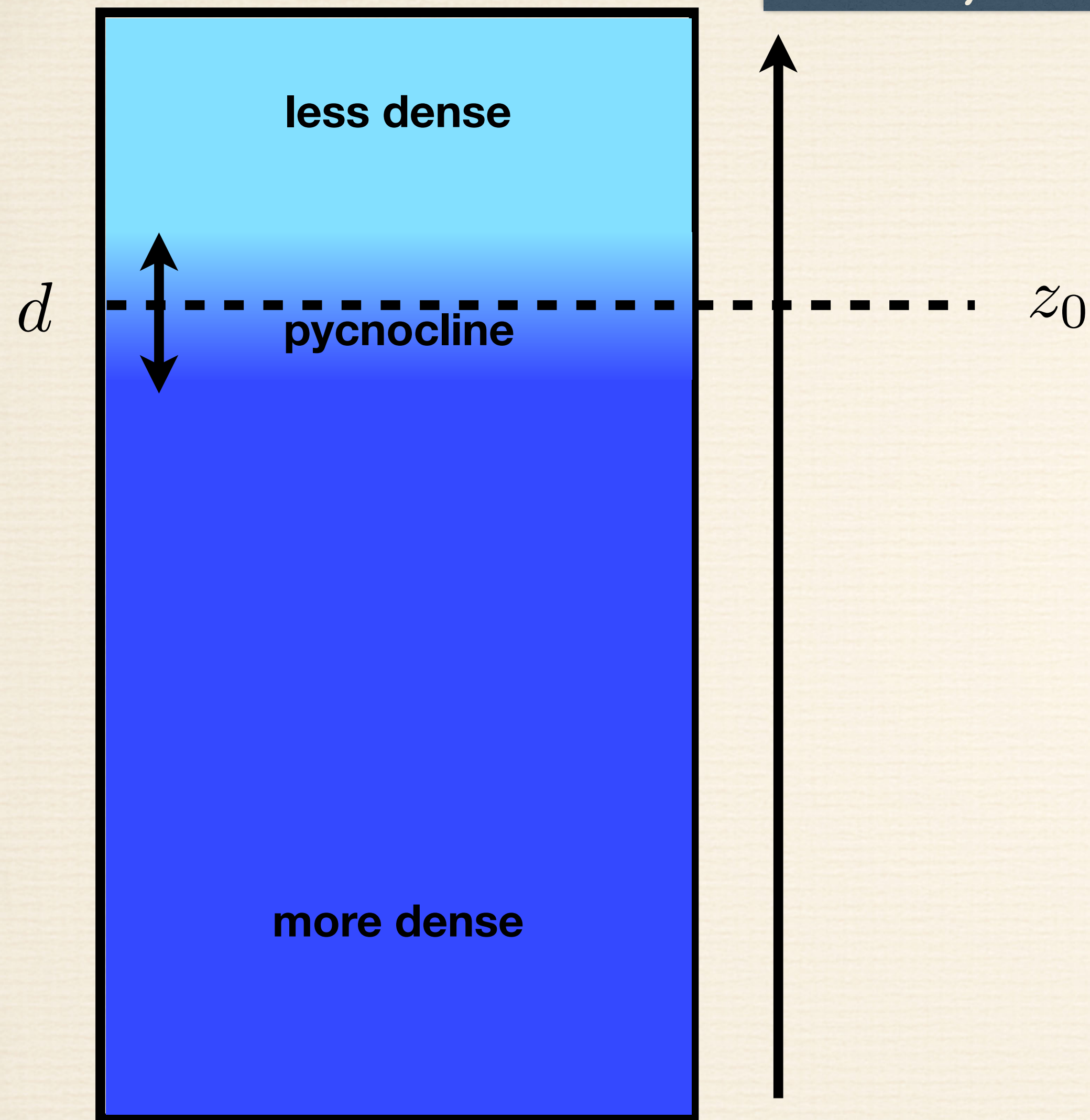
Stratified Euler Equations under the Boussinesq Approximation

Momentum	$\frac{D\vec{u}}{Dt}$	=	$-\frac{1}{\rho_0} \vec{\nabla} p$	-	$\frac{\rho}{\rho_0} g \hat{k}$	
Energy	$\frac{D\rho}{Dt}$	=	0			
Mass	$\vec{\nabla} \cdot \vec{u} = 0$					

Red boxes indicate places where dissipative processes like viscosity would appear in the Navier Stokes version

- The Boussinesq approximation makes the assumption that the density is constant EXCEPT in the buoyancy term (purple box).
- It can be applied to most naturally occurring fluids like air and water (a bit of a surprise, and a nice little write up for the lab book).
- There is no evolution equation for pressure.

Density Stratification: Basic Physics



$$\rho = \rho_0(\bar{\rho}(z) + \rho'(x, y, z, t))$$



- ❖ Reference density (think 1000 kg/m³ for water)
- ❖ Density stratification (see diagram on left)
- ❖ Density perturbation (e.g. due to wave motion)

- In Nature it is often observed that two layers of almost constant density are separated by a thin layer of rapid density change.
- The layer of density change is called the “pycnocline” (from Greek).
- A common analytical representation is given below (try varying the parameters for your lab book).

If denser fluid overlies less dense fluid get an instability

$$\bar{\rho}(z) = 1 - a \tanh [(z - z_0)/d]$$

Linearizing I

- ❖ We want to start with a density stratified, but non moving state: $\rho = \rho_0 \bar{\rho}(z), \vec{u} = 0$
- ❖ The stratified Euler equations under the Boussinesq approximation reduce to:
$$0 = -\frac{1}{\rho_0} \nabla p - \bar{\rho}(z) g \hat{k} \implies \frac{dp}{dz} = -\rho_0 \bar{\rho} g$$
 or the hydrostatic pressure relation.
- ❖ We next assume we can work in the x-z plane and assume all perturbations are small so that we can linearize the stratified Euler equations:

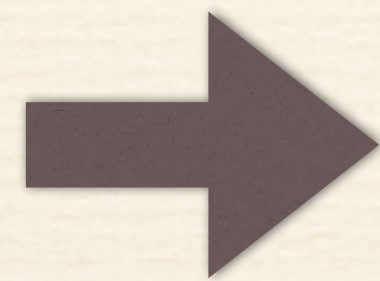
Details for lab book

$$\rho = \rho_0 \bar{\rho} + \epsilon \rho_0 \rho'(x, z, t)$$

$$\vec{u} = \vec{0} + \epsilon \vec{u}'(x, z, t)$$

$$p = p_H(z) + \epsilon p'(x, z, t)$$

linearize



$$u'_t = -p'_x / \rho_0 \quad w'_t = -p'_z / \rho_0 - g \rho'$$

$$u'_x + w'_z = 0$$

$$\rho'_t + w \bar{\rho}_z = 0$$

momentum

mass

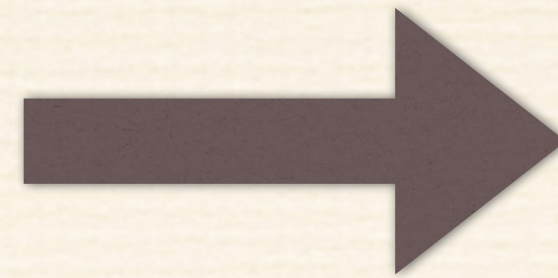
energy

Linearizing II

$$u'_t = -p'_x/\rho_0 \quad w'_t = -p'_z/\rho_0 - g\rho'$$

$$u'_x + w'_z = 0$$

$$\rho'_t + w\bar{\rho}_z = 0$$



$$u' = \psi_z, w' = -\psi_x$$

$$\omega'_t = g\rho'_x \implies \nabla^2 \psi_t = g\rho'_x$$

$$\rho'_t - \psi_x \bar{\rho}_z = 0$$

- ❖ We cross-differentiate the momentum equations with respect to x and z to form the linearized vorticity equation (notice there is a way to generate vorticity).
- ❖ We then use the stream function to simplify.
- ❖ Finally we can cross-differentiate the vorticity and density perturbation equations with respect to t and x and eliminate rho'.
- ❖ Introduce the so-called square of the buoyancy frequency $N^2(z)$ and get the final linearized equation governing perturbations from the density stratified state.

$$\nabla^2 \psi_{tt} = g\rho'_{xt} = \psi_{xx} g\bar{\rho}_z$$

$$N^2(z) := -g\bar{\rho}_z$$

$$\Downarrow \qquad \qquad \Downarrow \qquad \qquad \Downarrow$$

$$\nabla^2 \psi_{tt} + N^2(z) \psi_{xx} = 0$$

N^2 is measured

Solution strategies 1

- ❖ Let's take stock of where we have gotten to:
- ❖ First of all we have a basic state of a still fluid that is stably stratified. This is the “wave guide” for waves and GRAVITY is the restoring force.
- ❖ We then used the technique of linearization to derive equations for the perturbations from this basic state.
- ❖ By manipulating the equations we were able to get a single PDE. However, this PDE is not one of the ones you may have seen in PDE classes.
- ❖ However, if we imagine that what we want to study are waves propagating along the wave guide (in other words in the x direction) we should be able to use the wave ansatz from the last set of slides to make further progress:
$$\psi = \text{Re} [\exp(ik(x - ct))\phi(z)]$$
- ❖ Recall that here “ c ” is the phase speed.

Solution Strategies 2

$$\left. \begin{aligned} \nabla^2 \psi_{tt} + N^2(z) \psi_{xx} &= 0 \\ \psi &= \operatorname{Re} [\exp(ik(x - ct)) \phi(z)] \end{aligned} \right\} \begin{aligned} (-k^2 c^2)(\phi_{zz} - k^2 \phi) - k^2 N^2(z) \phi &= 0 \\ \phi_{zz} + \left(\frac{N^2}{c^2} - k^2 \right) \phi &= 0 \end{aligned}$$

- ❖ If we consider a domain with $z=0$ being the bottom and $z=H$ the surface, the simplest boundary conditions are $w'=0$, or no flux through the top and bottom.
- ❖ Since $w' = -\psi_x$ the wave ansatz tells us that: $\phi(0) = \phi(H) = 0$.
- ❖ Thus the whole problem of describing waves amounts to solving an ODE for ϕ and c . Remember we think of the wave number k as being specified.
- ❖ If the fluid is “stably stratified” so that less dense fluid sits over top denser fluid (this amounts to $N^2 \geq 0$) c is real.

An Important Special Case I

$$\phi_{zz} + \left(\frac{N^2}{c^2} - k^2 \right) \phi = 0, \phi(0) = \phi(H) = 0$$

- ❖ In general the problem for ϕ and c must be solved numerically. But there is one special case that can be solved by hand.
- ❖ This is the case of a linear stratification: $\bar{\rho}(z) = 1 - \alpha z \implies N^2(z) = N_0^2$ **Constant**
- ❖ The ODE can be written as: $\phi_{zz} + m^2 \phi = 0$ where $m^2 = N_0^2/c^2 - k^2$
- ❖ When $m^2 > 0$ this is a simple harmonic oscillator equation and we can just guess a solution that satisfies the boundary condition at $z=0$: $\phi(z) = \sin(mz)$
- ❖ To satisfy the boundary condition at $z=H$ we need: $\sin(mH) = 0 \implies mH = n\pi$ where $n = 1, 2, \dots$ which tells us how many times ϕ oscillates between $z=0, H$.

An Important Special Case II

$$m^2 = \frac{n^2 \pi^2}{H^2} \implies \frac{N_0^2}{c^2} = \frac{n^2 \pi^2}{H^2} + k^2 \implies c^2 = \frac{N_0^2 H^2}{n^2 \pi^2 + k^2 H^2} \implies c_{\pm} = \pm \frac{N_0 H}{\sqrt{n^2 \pi^2 + k^2 H^2}}$$

- ❖ A bit of algebra allows us to solve for the phase speed.
- ❖ You can see that the phase speed decreases as n increases.
- ❖ You can also see that the phase speed decreases as k increases, or as the wavelength gets shorter.
- ❖ This means these waves are **dispersive**.
- ❖ N_0 and H are physical parameters assumed to be measured.

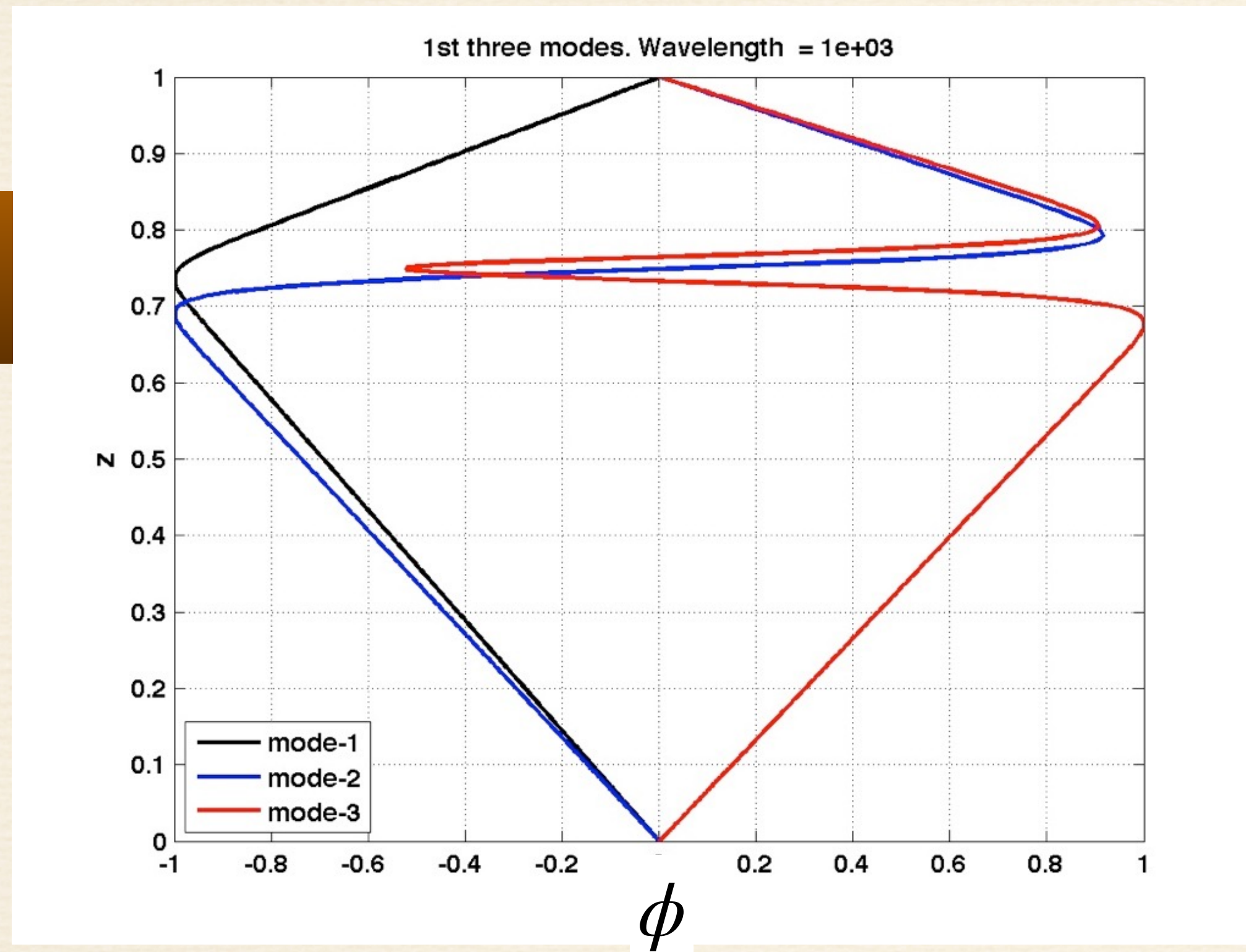
An Important Special Case III

$$c = \frac{N_0 H}{\sqrt{n^2 \pi^2 + k^2 H^2}} \implies \sigma = \frac{N_0 H k}{\sqrt{n^2 \pi^2 + k^2 H^2}} \implies c_g = \frac{N_0 H n^2 \pi^2}{(n^2 \pi^2 + k^2 H^2)^{3/2}}$$

- ❖ A bit of algebra allows us to solve for the frequency $\sigma(k)$ and using the definition $c_g = \frac{d\sigma}{dk}$ for the group velocity.
- ❖ You can analyze the group velocity for your lab book.
- ❖ The linearly stratified case is useful mathematically, but not representative of Nature.
- ❖ It also has the somewhat strange property that ϕ does not change as the wavelength changes.

Linear Waves: sample numerical solutions, vertical structure I

pycnocline
located here



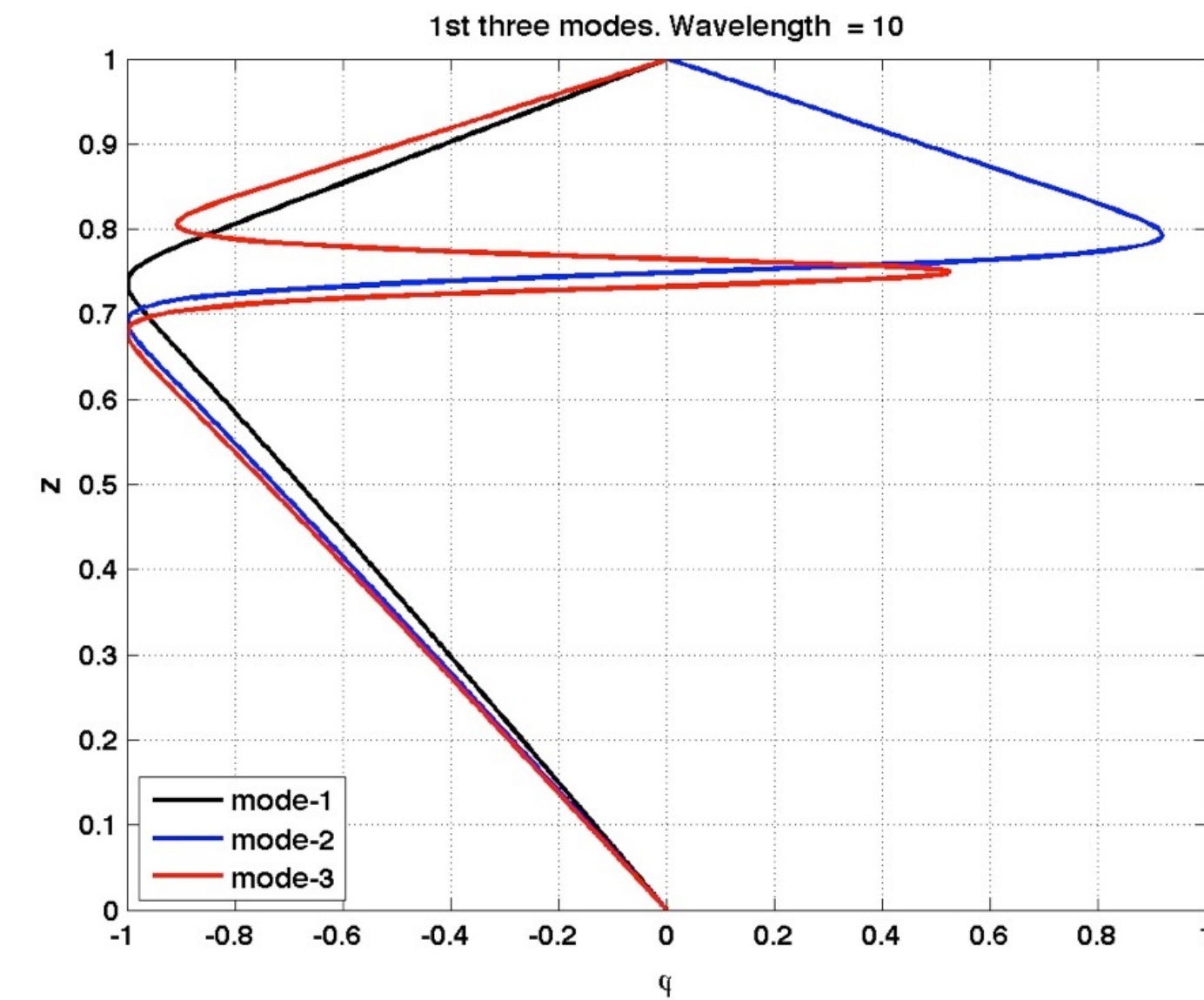
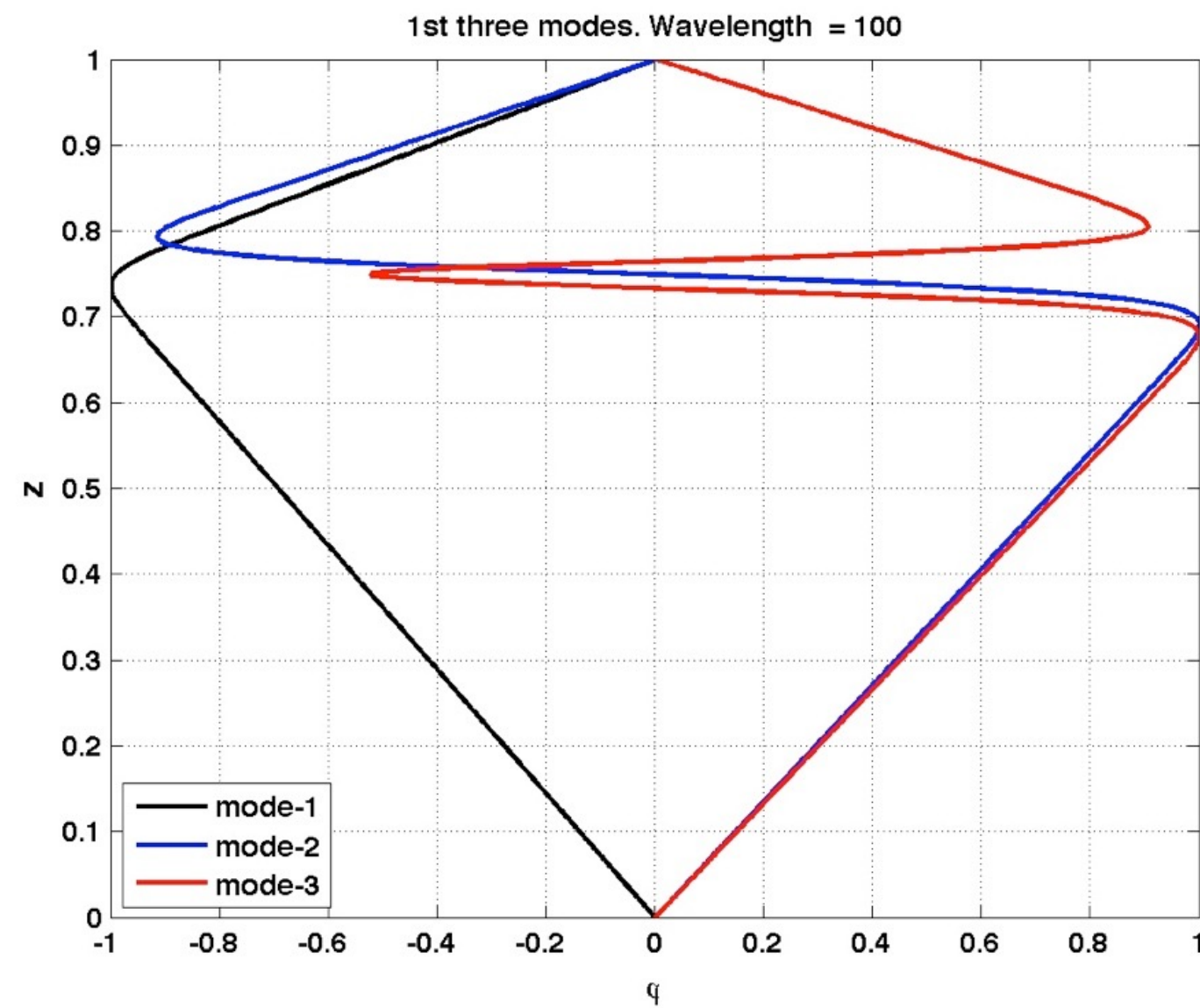
$$\bar{\rho}(z) = 1 - a \tanh [(z - z_0)/d]$$

$$z_0 = 0.75, d = 0.05$$

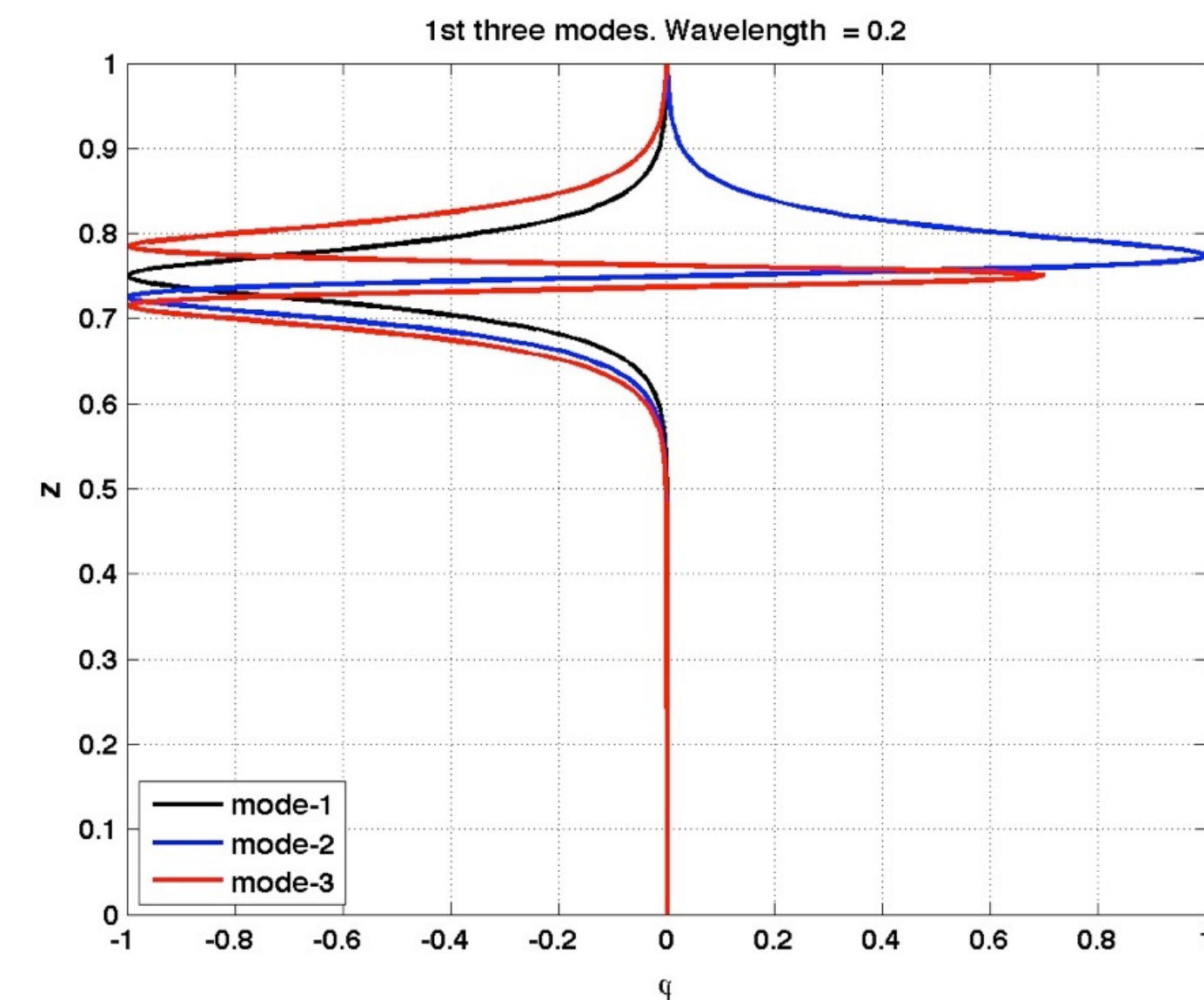
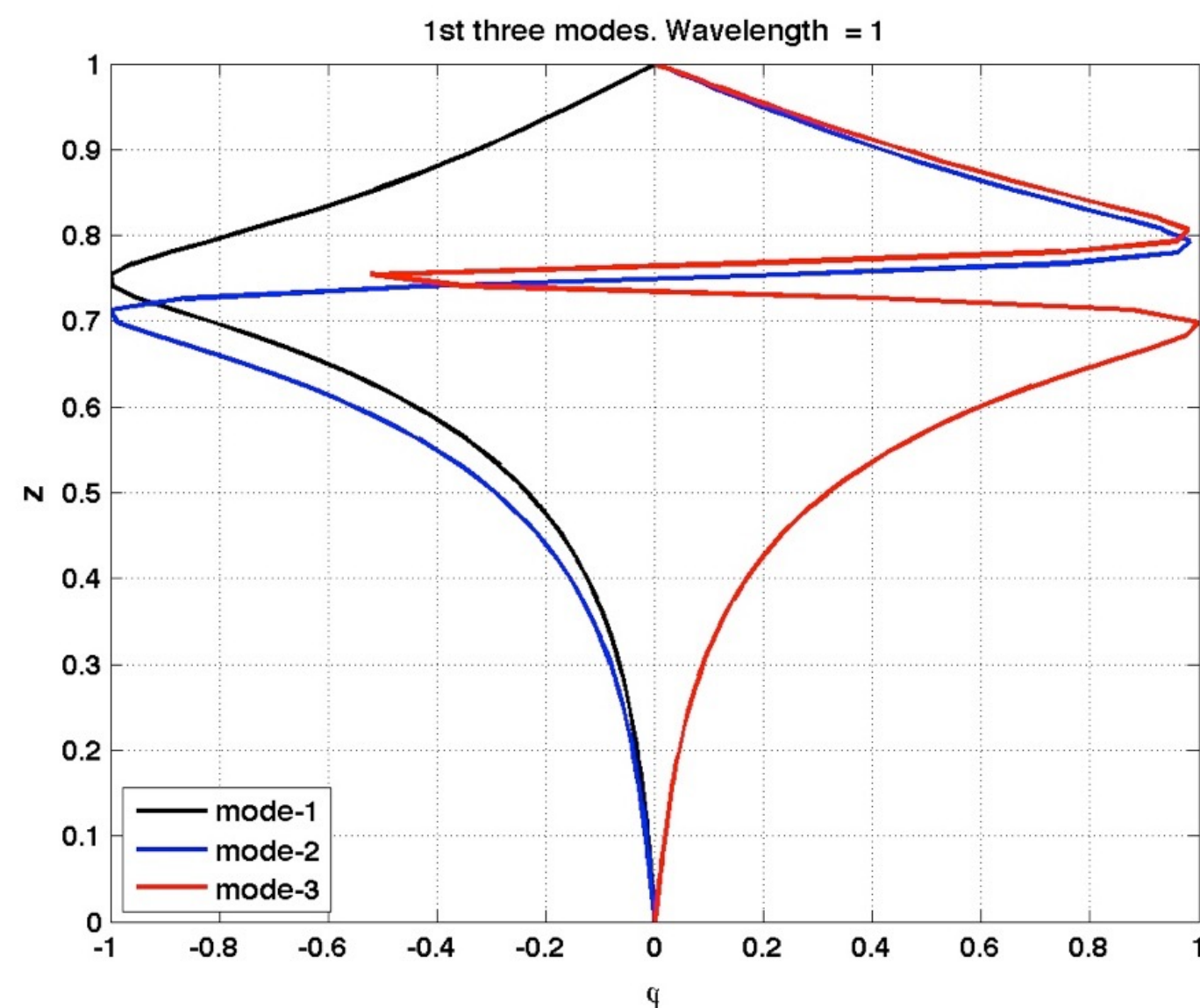
- ❖ I solve the ODE eigenvalue problem for ϕ and c in Matlab (am happy to share the solver).
- ❖ Mode number is defined as the number of zero crossings in the interior, plus one. So $n=1$ means mode 1, $n=2$ means mode 2 and so on
- ❖ The first three modes are shown on the left
- ❖ You can see that mode 1 and 2 behave about as you would expect.
- ❖ Mode 3 is a bit unusual with a very rapid change near $z=0.75$
- ❖ On the next slide I show you what varying k changes.

Linear Waves: sample numerical solutions, vertical structure I

mode
number
is
n

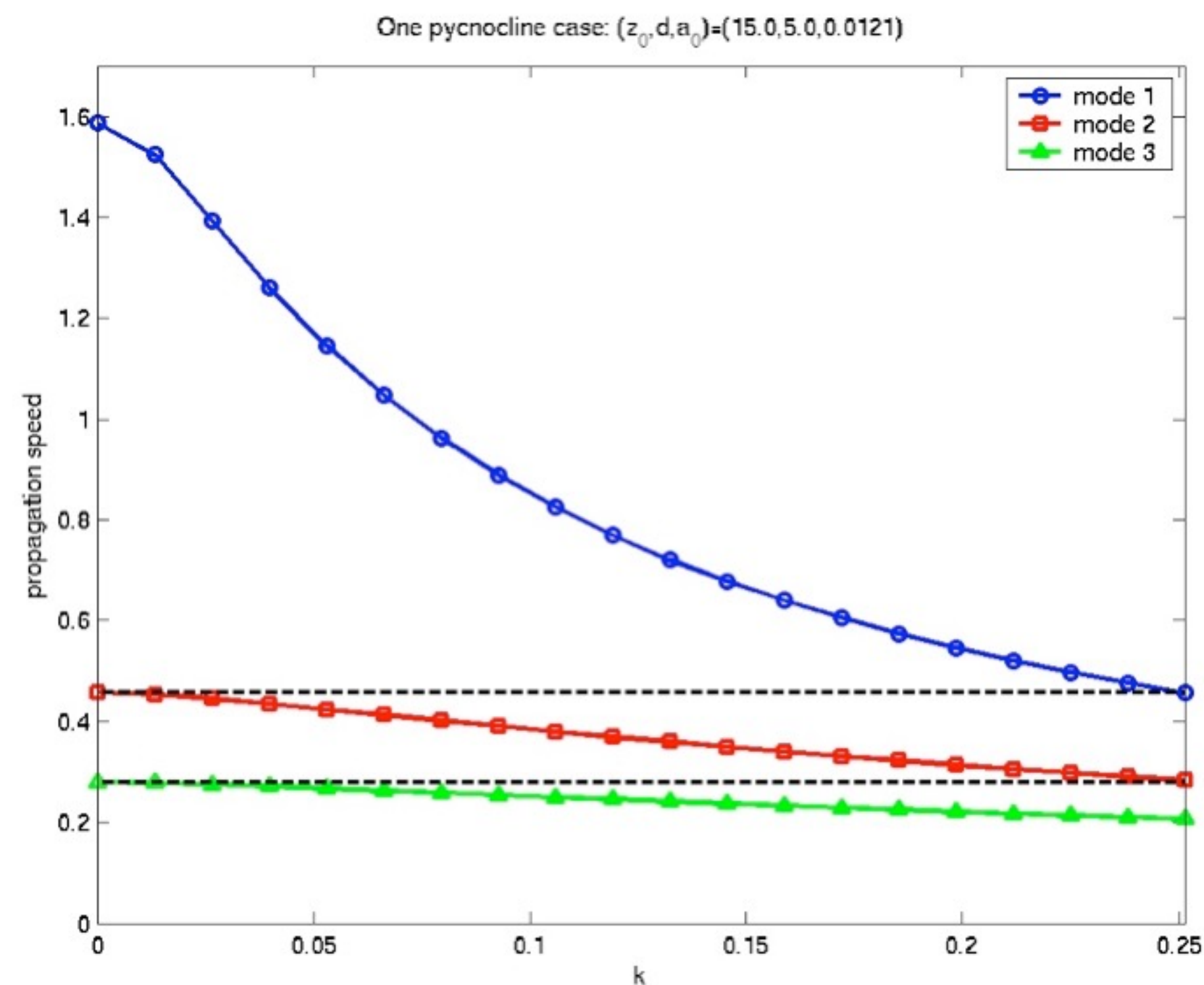


$$k = \frac{2\pi}{\text{wavelength}}$$



Shorter waves
are more
trapped near the
pycnocline

Linear Waves: sample numerical solutions, phase speed



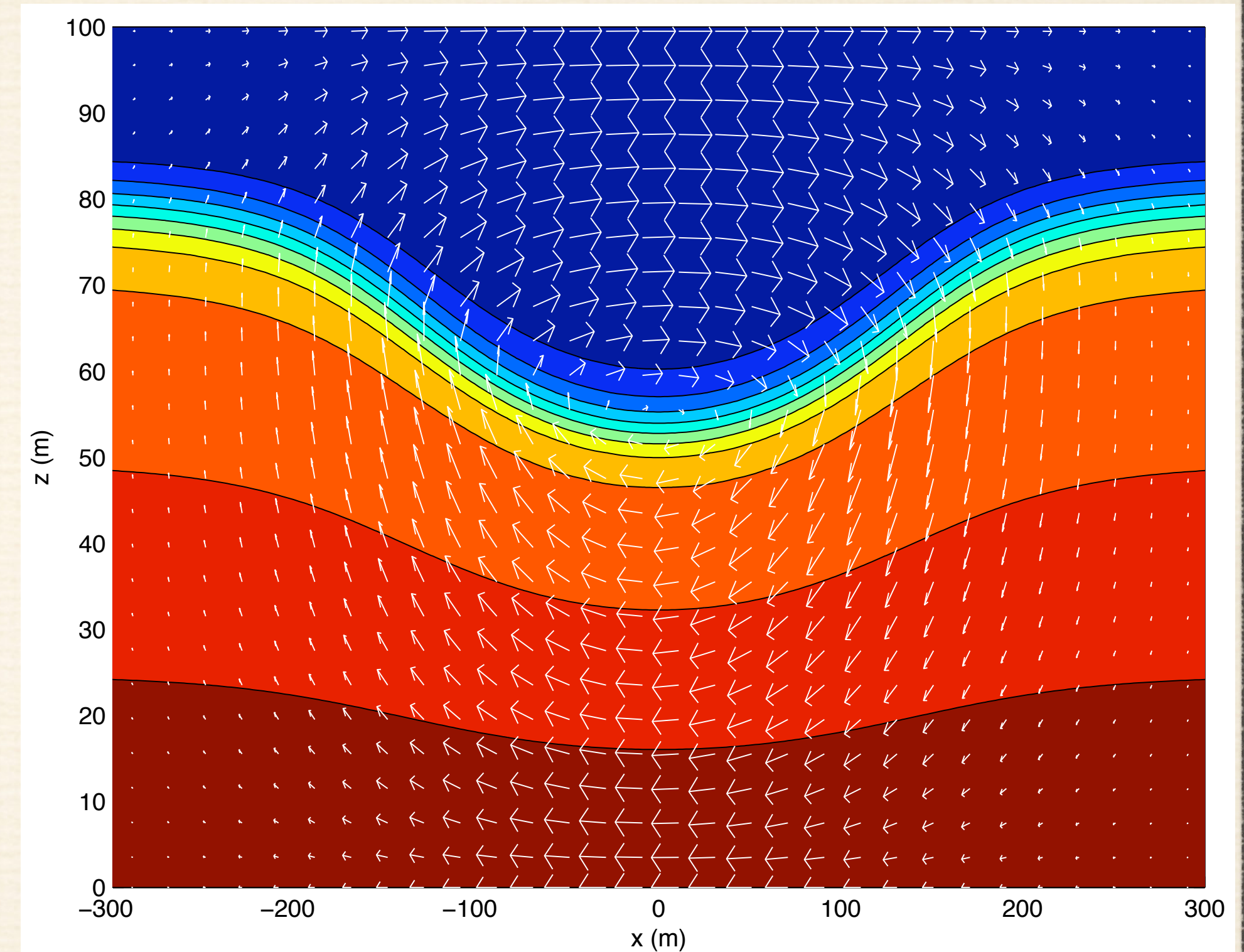
Long
waves

Short
waves

- ❖ I next solve the ODE eigenvalue problem repeatedly varying the wavenumber k .
- ❖ I then plot the phase speed c as a function of k .
- ❖ Long mode-1 waves propagate the fastest.
- ❖ Long mode-2 waves propagate at the same speed as short mode 1 waves.
- ❖ The full solution for a physical field is computed iteratively.
- ❖ For example $u = \psi_z = \cos(k(x - c(k)t))\phi_z(z)$ and $w = -\psi_x = k \sin(k(x - c(k)t))\phi(z)$

Conclusions

- ❖ Stable stratification in which density decreases with z provides a wave guide for internal gravity waves.
- ❖ Internal waves are dispersive, and as such are NOT a solution of the classical wave equation.
- ❖ Internal waves are dispersive, and as such energy does not move at the same speed as individual crests.
- ❖ Of course linear theory is only a start as the wave on the right, which is an exact solution of the FULL stratified Euler equations, shows.



Internal solitary wave, density field is shaded and the velocity field is shown in white. Wave moves left to right.

A “physics” derivation of $N^2(z)$

- ❖ Consider a fluid particle initially found at $z = z_0$, so that its density is $\bar{\rho}(z_0)$
- ❖ Now assume that the particle is displaced to $z = z_0 + \Delta z(t)$ and released.
- ❖ The only force per unit area will be the buoyancy based on the difference in density of the particle $\bar{\rho}(z_0)$ and the local density $\bar{\rho}(z_0 + \Delta z)$. Newton’s second law then reads:
- ❖ $\rho_0 \frac{d^2 \Delta z}{dt^2} = -g (\bar{\rho}(z_0) - \bar{\rho}(z_0 + \Delta z))$ where I used the Boussinesq approximation to set the density on the LHS to the reference density.
- ❖ Now Taylor expand to get: $\rho_0 \frac{d^2 \Delta z}{dt^2} \approx -g \bar{\rho}'(z_0) \Delta z$ which can be rearranged to look like the simple harmonic oscillator:
- ❖ $\frac{d^2 \Delta z}{dt^2} + N^2 \Delta z = 0$ where $N^2(z_0) = -\frac{g}{\rho_0} \bar{\rho}'(z_0)$ and has dimensions of frequency squared hence the term **buoyancy frequency**.